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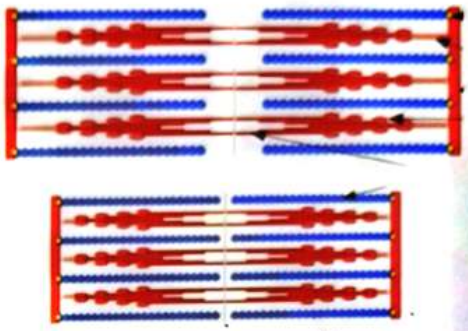
----- Active space -----

PROPERTIES OF SKELETAL MUSCLES

Length Tension Relationship

00:02:00

Contraction of a muscle fibre
- Shortening of muscle fibre → Change in length
- Generation of force → Change in tension

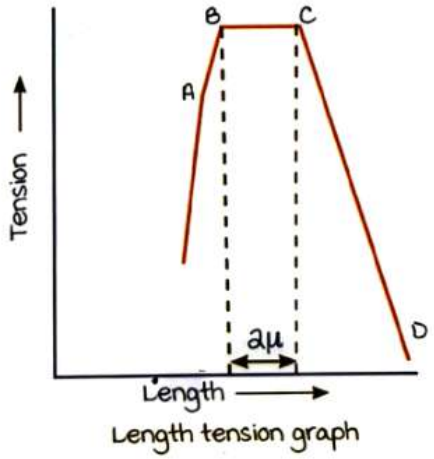


Actin myosin overlap

LENGTH TENSION RELATIONSHIP GRAPH :

Length is compared to active tension.

	Actin myosin overlap	Tension generated	Image
Point D	Absent	No tension	Myosin, Actin, No overlap
Point B & C	Optimal	Maximum tension (Tmax)	Optimal overlap
Point A	Present, but not optimal	Less tension	Not optimal overlap



Length tension graph

Types of Tension & Length

00:06:44

Types of tension
- Active
- Passive
- Total

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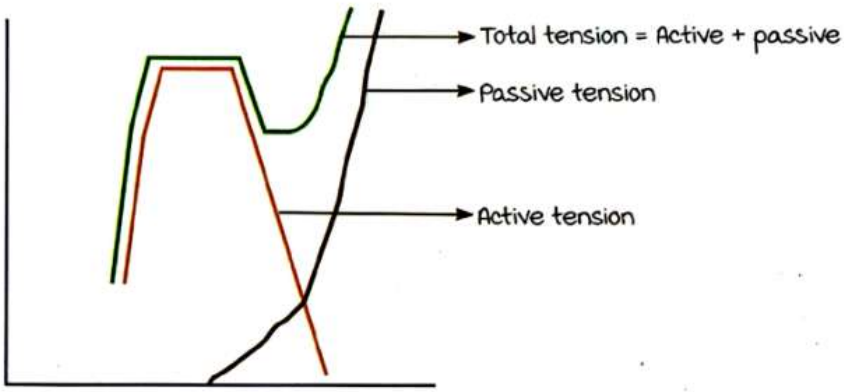
Active v/s passive tension :

----- Active space -----

	Active	Passive
Tension generated	When muscle is stimulated	When muscle is not stimulated
Proteins involved	Actin myosin overlap ↓ Cross bridge formation ↓ Tension	Titin ↓ Passive stretching/ elasticity

Factors affecting active tension :

- Cross bridge formation.
- Diameter of muscle fibre.
- Length of muscle fibre.
- AP frequency.
- No. of motor units recruited.



Types of tension

Types of length

- Optimal/resting length : Maximum tension generated by all body muscles
- Equilibrium length : Muscles are cut from bony attachments

Types of muscle fibres

- Type I : Small & red
- Type II
  - Type IIa : Large & red
  - Type IIb : Large & white

Characteristics	Type I fibres	Type II fibres	
		Type II a	Type II b
Color	Red	Red	White
myoglobin content	(+) High	(+) High	(-) Absent
Diameter	Small	Large	Large
Myosin ATPase (Contraction)	Slow	Fast	Fast
Metabolism O2 storage Pathway to generate ATP lactate	(+) Aerobic pathway (-)	(+) Aerobic or anaerobic pathway (Glycolysis) (+)	(-) Anaerobic pathway (Glycolysis) (+) Pain : muscle fatigue

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----- Active space -----

Characteristics	Type I fibres	Type II fibres	
		Type II a	Type II b
Fatigue	Fatigue resistant	Fatigue resistant	Fatigue prone
motor units innervation	Slow fatigue resistant	Fast fatigue resistant	Fast fatiguable
Recruitment order	First d/t small diameter (Size principle)	Second	Last
Activities best suited	Sitting maintaining posture	Standing	Running

Motor Unit

00:30:25

Single α motor neuron + muscle fibre it innervates → motor unit.

Fine precise movements :

Eg : Extraocular muscles.

motor unit innervates → 3-6 muscle fibres.

Coarse movements :

Eg : Leg/back muscles.

motor unit innervates → 600 muscle fibres.

Muscle Heat

00:33:45

During contraction & relaxation → muscle produces heat.

Types of heat :

	Resting heat	Contraction		Relaxation	
		Activation heat	Shortening heat	Relaxation proper	Recovery heat
Due to	Basal metabolic rate	Activation of actin & myosin	Overlap of actin & myosin	Isotonic contraction No tension change	muscle regains original length

Note : Examples of ↑ resting heat.

- Lean body.
- After exercise (weight loss).

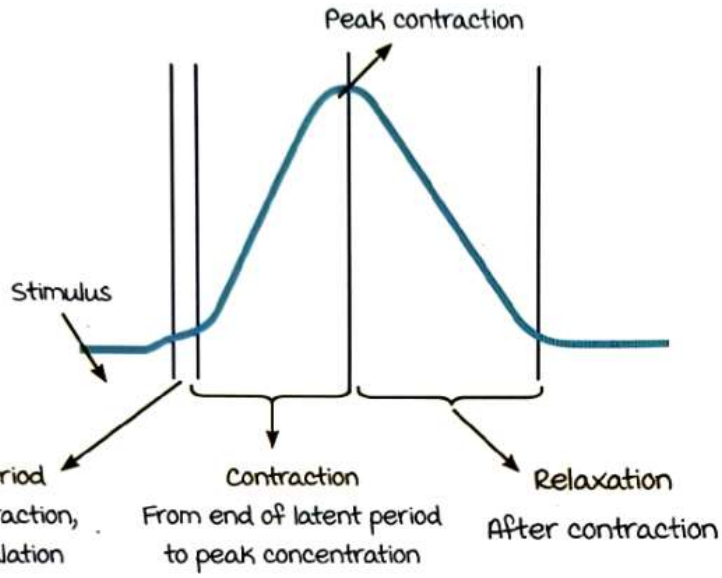
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Stimulation of a Muscle

00:38:56 ----- Active space -----

Single muscle twitch :



INCOMPLETE TETANUS :

↑ frequency of stimulus → ↑ frequency of contraction.
Incomplete relaxation.

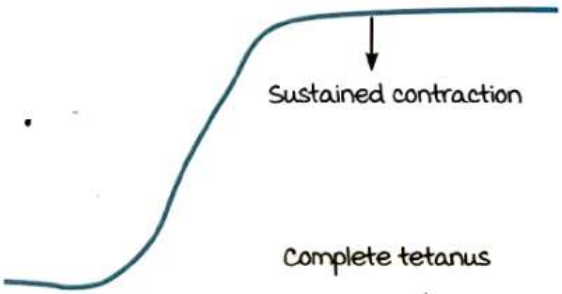


COMPLETE TETANUS :

↑↑↑ frequency of stimulus → Sustained contraction & no relaxation.

D/t Ca^{2+} ion accumulation.

Eg : Tetanospasmin (In tetanus d/t Clostridium tetani)
↓
Spastic contracted state
↓
Body bends like a bow



Tetanising frequency (T_f) :

Frequency of stimulus at which complete tetanus is produced.

$$T_f = \frac{1}{\text{Contraction period}}$$

Fibrillation v/s fasciculation

00:46:13

	Fibrillation	Fasciculation
No. of muscle fibres	Single	multiple
No. of motor units	Single (α motor neuron or its axons)	multiple

Feedback